

EXPERIENCE

- **Motorola Solutions** Chicago, IL
Software Engineer Intern *Jun. 2024 - Present*
 - **Continuous Integration:** Configured **Jenkins pipeline for continuous integration** of hardware security modules. Programmed **shell scripts** for remote execution on Windows and Linux-based systems interfacing with Python unit test infrastructure.
 - **GenAI Features:** Created text suggestion and conversation summary features with **Java + Kotlin for Android** messaging app deployed on first-responder radios. Used Qualcomm **ML compilers** for conversion and quantization of lightweight transformer models in memory-constrained environments for **on-device GPU acceleration**.
 - **Unit Testing:** Creation/debugging of unit tests in MockK and Python unittest.
- **LG Electronics** Cleveland, OH
Software Engineer Intern *Aug. 2023 - Dec. 2023*
 - **Web Scraping:** Created web-scraping tool using **Python and BeautifulSoup** for obtaining product reviews from multiple product retail sites; after filtering, the tool collected **150,000 product reviews**.
 - **Data Analysis:** Storage and querying of product reviews using SQLite. Natural Language ToolKit for sentiment analysis and named entity recognition. Visualization with Python Matplotlib and Microsoft Excel.
- **Case School of Engineering** Cleveland, OH
Head Teaching Assistant *Aug. 2023 - May. 2025*
 - **Operating Systems:** Created and taught lessons for in-person lecture; requires clear understanding of Linux CLI, C programming language, and operating systems topics such as scheduling, virtual memory, and system architecture.
 - **Algorithms:** Created assignments + solutions with extensive use of LaTeX, and held additional instruction sessions; requires understanding of dynamic programming, divide-and-conquer algorithms, mathematical formalism + proofs, and graph algorithms.

PROJECTS

- **Photo Distribution Mobile App:** Multi-platform mobile app for efficiently routing photos from group events via facial recognition.
Tools/Languages: React-Native, Expo for frontend; Nest.js, PostgreSQL for backend; AWS Rekognition, S3, EC2 and Docker.
- **Automated Recruiting Tool:** Transformed manual tracking of 500 recruits per year to completely automated tracking. The tracking tool fetches updated rankings, analyzes recruits' ranking movement, and sends automated text messages.
Tools/Languages: Python, SQLite, BeautifulSoup, Pandas, Google Sheet API, AppleScript.
- **Compiler Front-End:** Designed a compiler front end for subset of the Lisp programming language. The grammar allows for homogeneous lists, dynamic typing, and pure functions.
Tools/Languages: C++, Flex, Bison, LLVM, Shell Scripting.

PROGRAMMING SKILLS

- **Languages:** Python, Java, Kotlin, C, Shell Scripting, JavaScript, SQL
- **Technologies:** AWS, React Native, Expo, Nest.js, Jenkins, Docker, Scikit-learn, JUnit, MockK
- **Tools:** Git, Jira, LaTeX, Vim

EDUCATION

- **Case Western Reserve University** Cleveland, OH
Bachelors of Science in Computer Science & Math; GPA: 4.00 *Aug. 2022 - Present*
 - **Men's Varsity Tennis Team:** 2023 NCAA D3 Team National Champions and ranked Top 5 nationally since joined program. Committing 25 hours per week to practice, lifting, and competitions while maintaining a full academic schedule.